

Figures

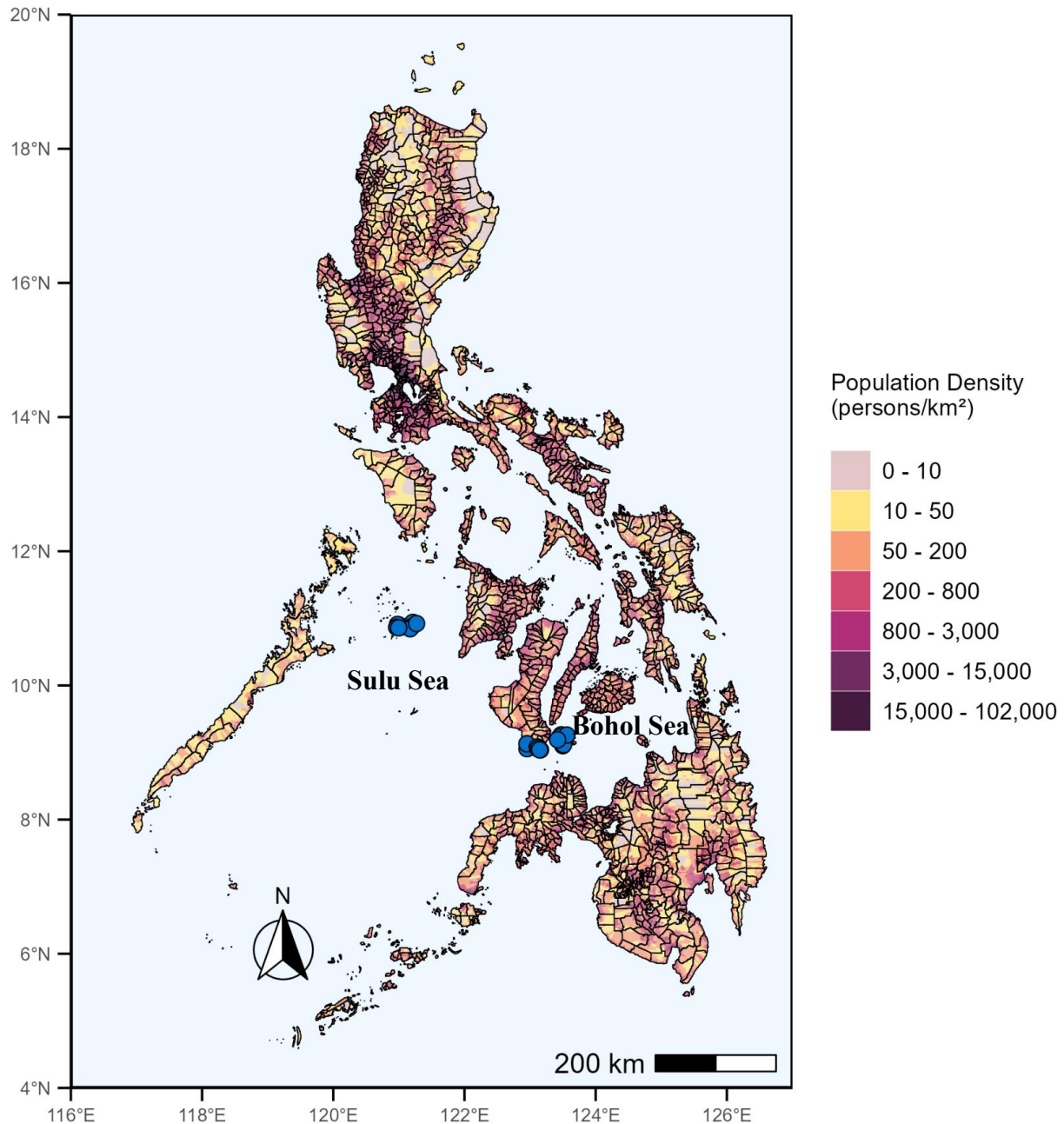


Figure 1. Map of 21 fish community sampling stations (blue circles) in the Sulu and Bohol Seas of the Visayas Region in the Philippines. The Smithsonian Institution survey occurred in 1978 and 1979. These stations were duplicated during the Silliman University survey in 2019 and 2022. Black lines indicate municipality borders. Color-coding corresponds to 2020 population density (persons/km²), data from the Gridded Population of the World collection (GPWv4).

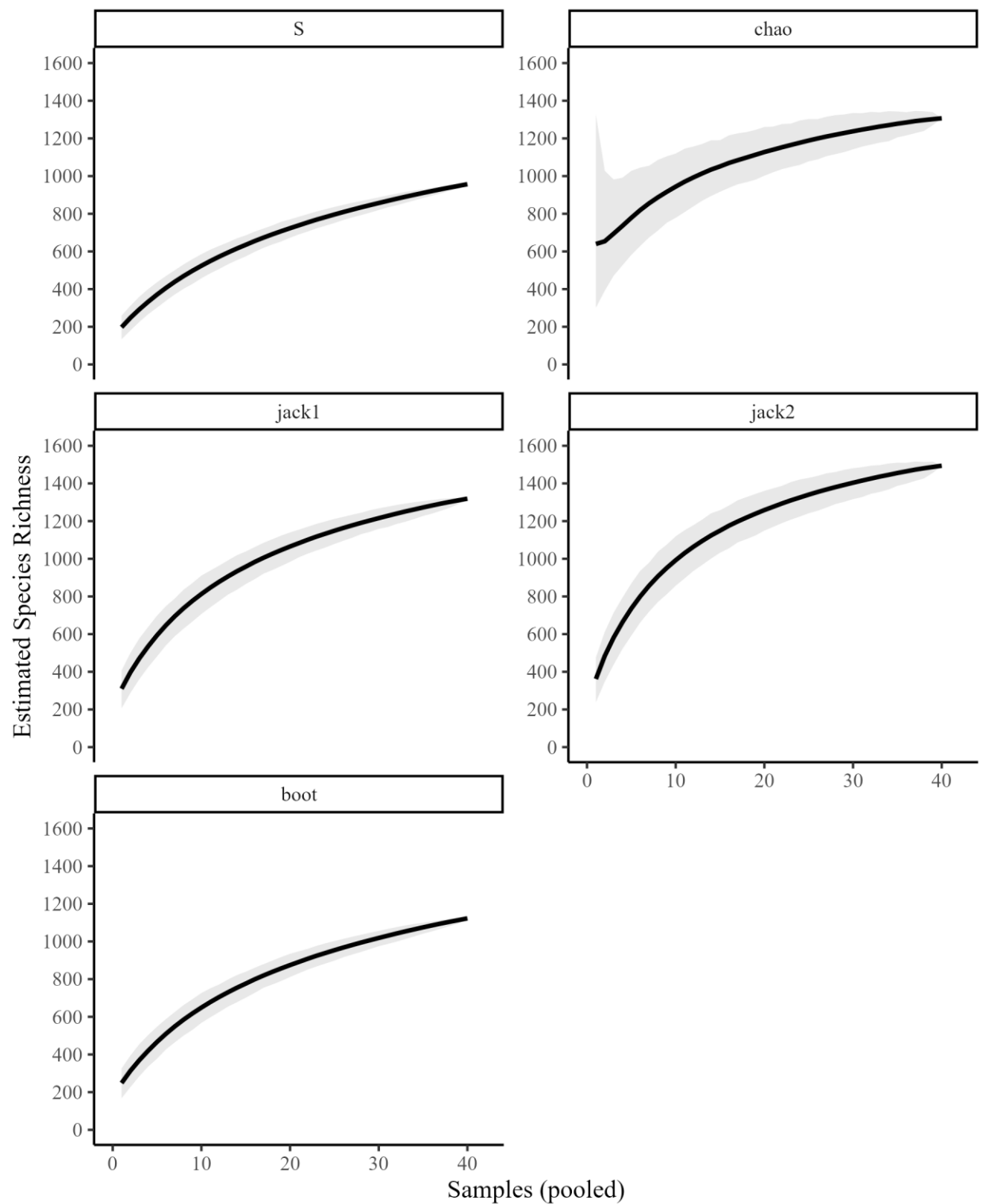


Figure 2. Sample-based species accumulation curves for reef fish communities surveyed in the Sulu and Bohol Seas during both survey eras (SI 1978/79, SU 2019/22). Curves show the estimated species richness as a function of the number of pooled samples, calculated using the *poolaccum* function in the *vegan* package with 999 random permutations. Each panel displays a different species richness estimator: S = observed richness, chao = Chao, jack1 = first-order jackknife, jack2 = second-order jackknife, boot = bootstrap. The solid lines indicate the mean estimate; shaded ribbons indicate 95% confidence intervals based on the empirical quantiles across permutations.

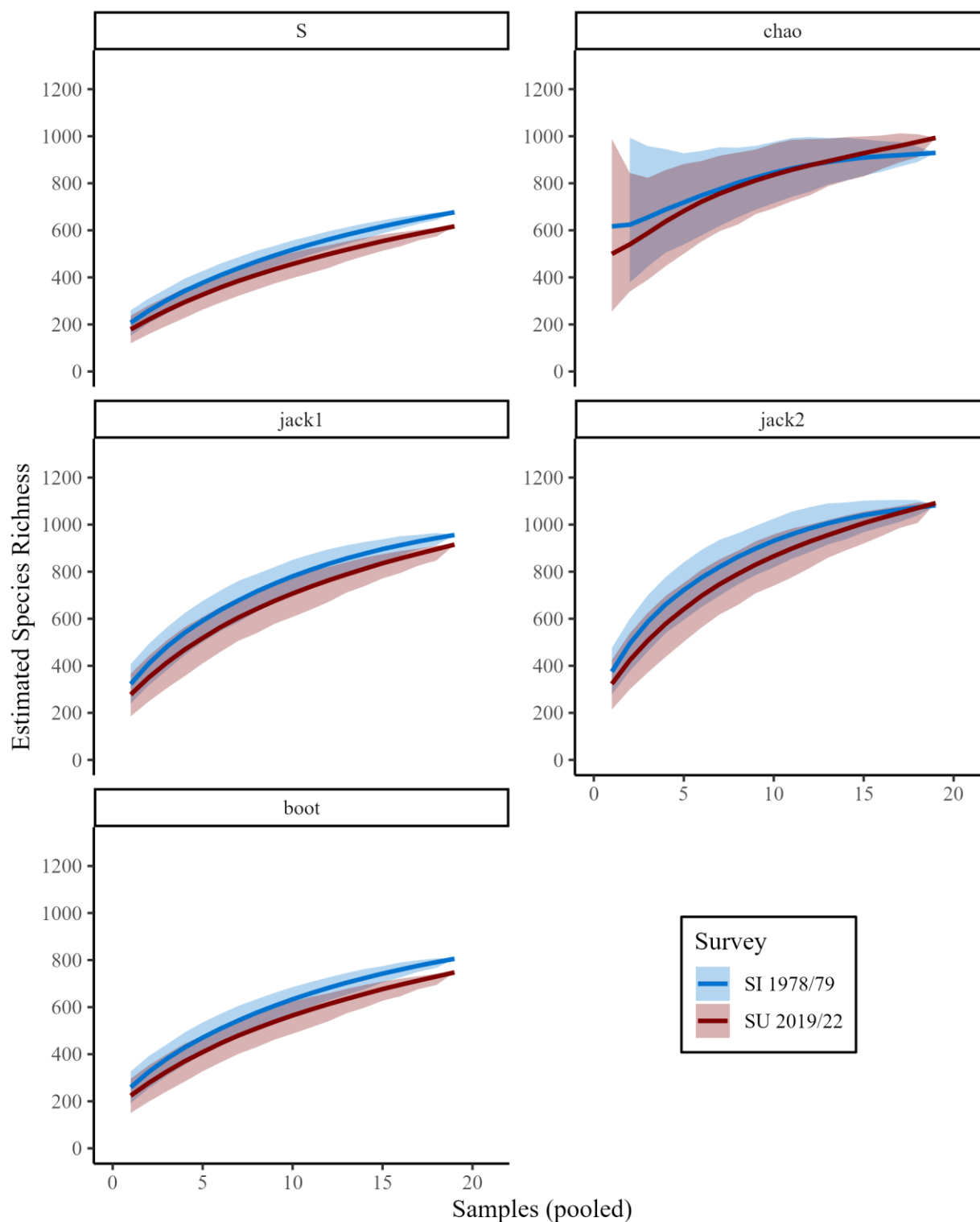


Figure 3. Sample-based species accumulation curves for reef fish communities surveyed in the Sulu and Bohol Seas during two survey eras (blue = SI 1978/79, red = SU 2019/22). Curves show the estimated species richness as a function of the number of pooled samples, calculated using the *poolaccum* function in the *vegan* package with 999 random permutations. Each panel displays a different species richness estimator: S = observed richness, chao = Chao1, jack1 = first-order jackknife, jack2 = second-order jackknife, boot = bootstrap. The solid lines indicate the mean estimate; shaded ribbons indicate 95% confidence intervals based on the empirical quantiles across permutations.

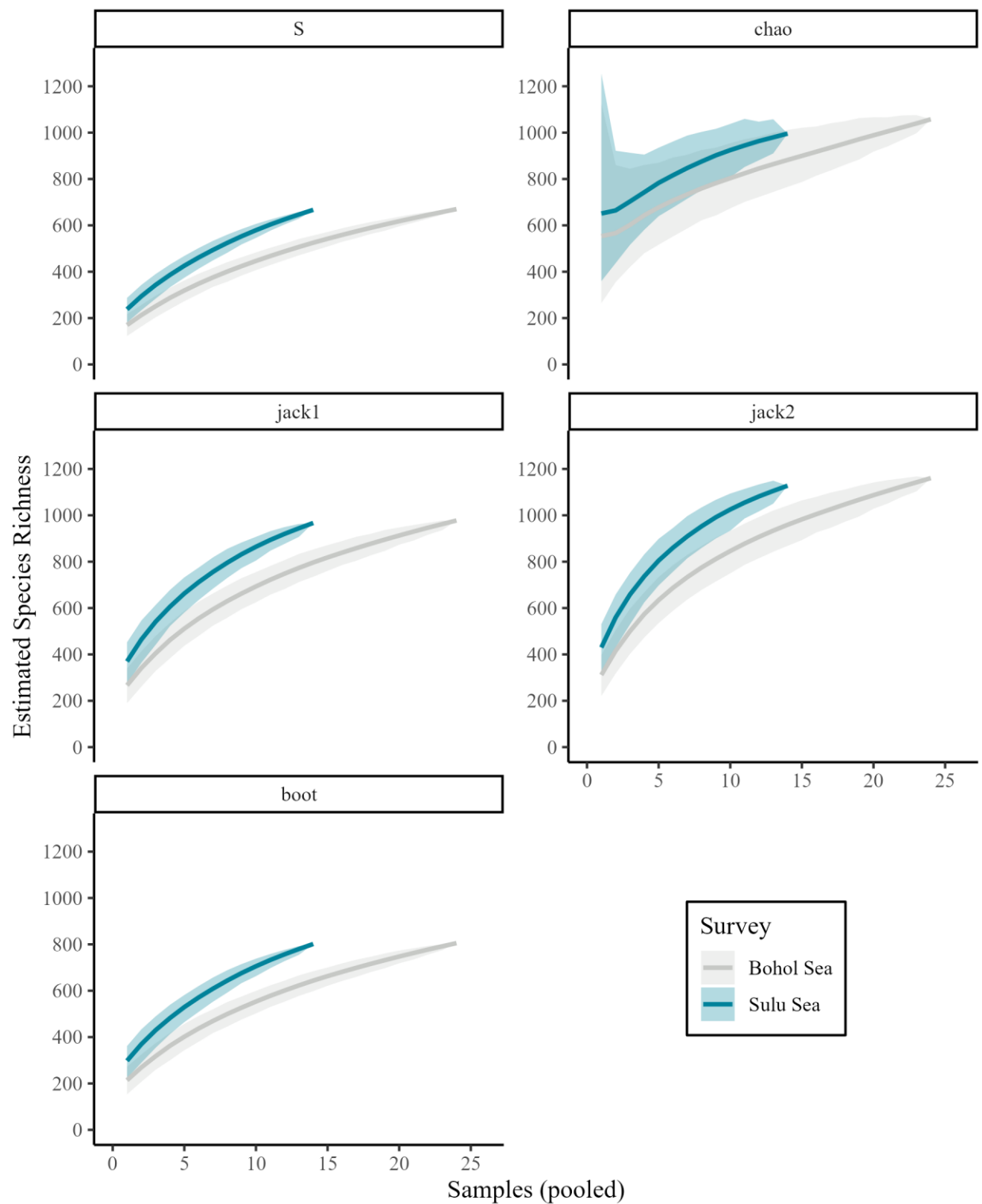


Figure 4. Sample-based species accumulation curves for reef fish communities surveyed in the Sulu (teal) and Bohol (grey) Seas during both survey eras (SI 1978/79, SU 2019/22). Curves show the estimated species richness as a function of the number of pooled samples, calculated using the *poolaccum* function in the *vegan* package with 999 random permutations. Each panel displays a different species richness estimator: S = observed richness, chao = Chao1, jack1 = first-order jackknife, jack2 = second-order jackknife, boot = bootstrap. The solid lines indicate the mean estimate; shaded ribbons indicate 95% confidence intervals based on the empirical quantiles across permutations.

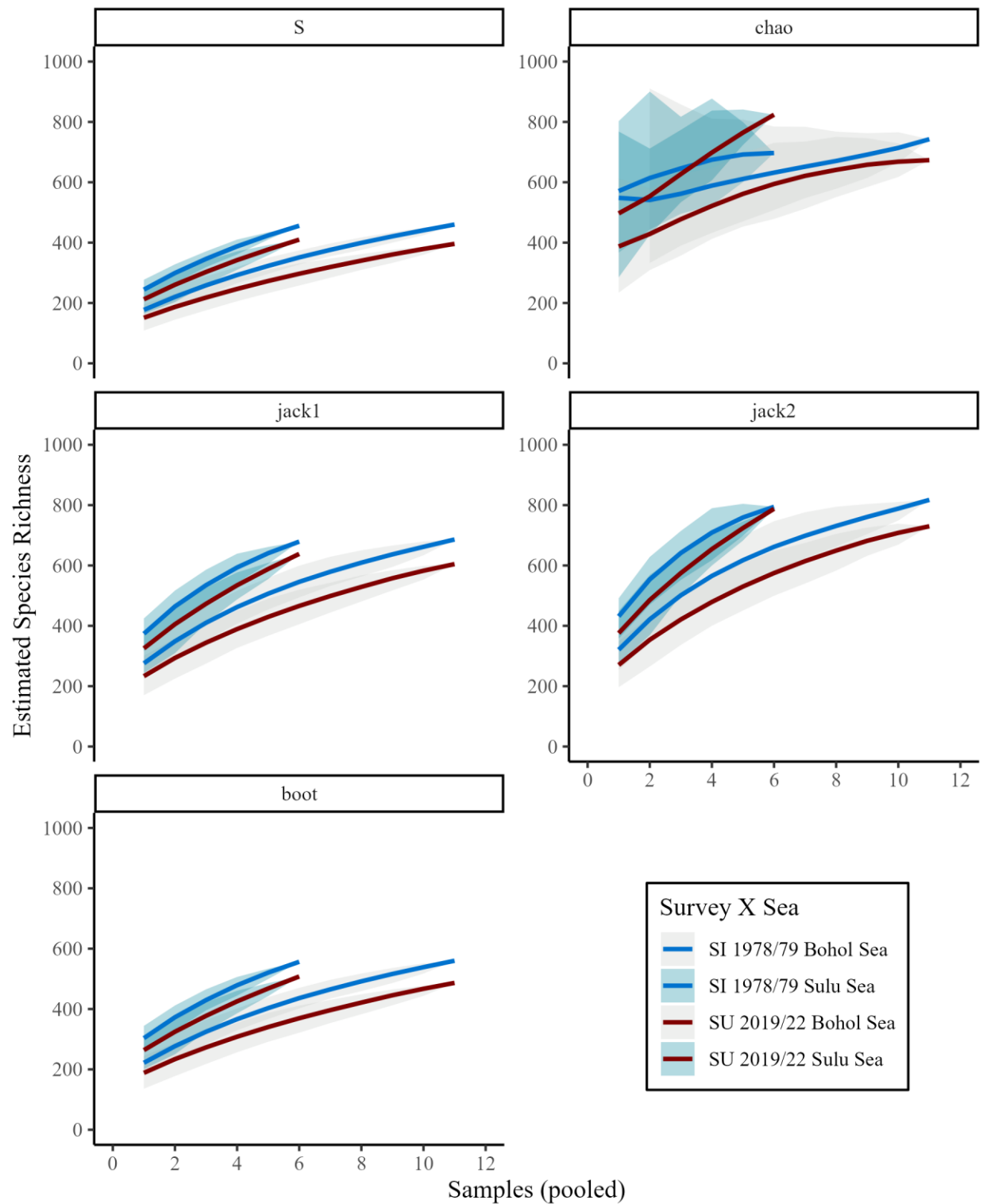


Figure 5. Sample-based species accumulation curves for reef fish communities surveyed in the Sulu (teal ribbon) and Bohol (grey ribbon) Seas during two survey eras (blue line = SI 1978/79, red line = SU 2019/22). Curves show the estimated species richness as a function of the number of pooled samples, calculated using the *poolaccum* function in the *vegan* package with 999 random permutations. Each panel displays a different species richness estimator: S = observed richness, chao = Chao1, jack1 = first-order jackknife, jack2 = second-order jackknife, boot = bootstrap. The solid lines indicate the mean estimate; shaded ribbons indicate 95% confidence intervals based on the empirical quantiles across permutations.

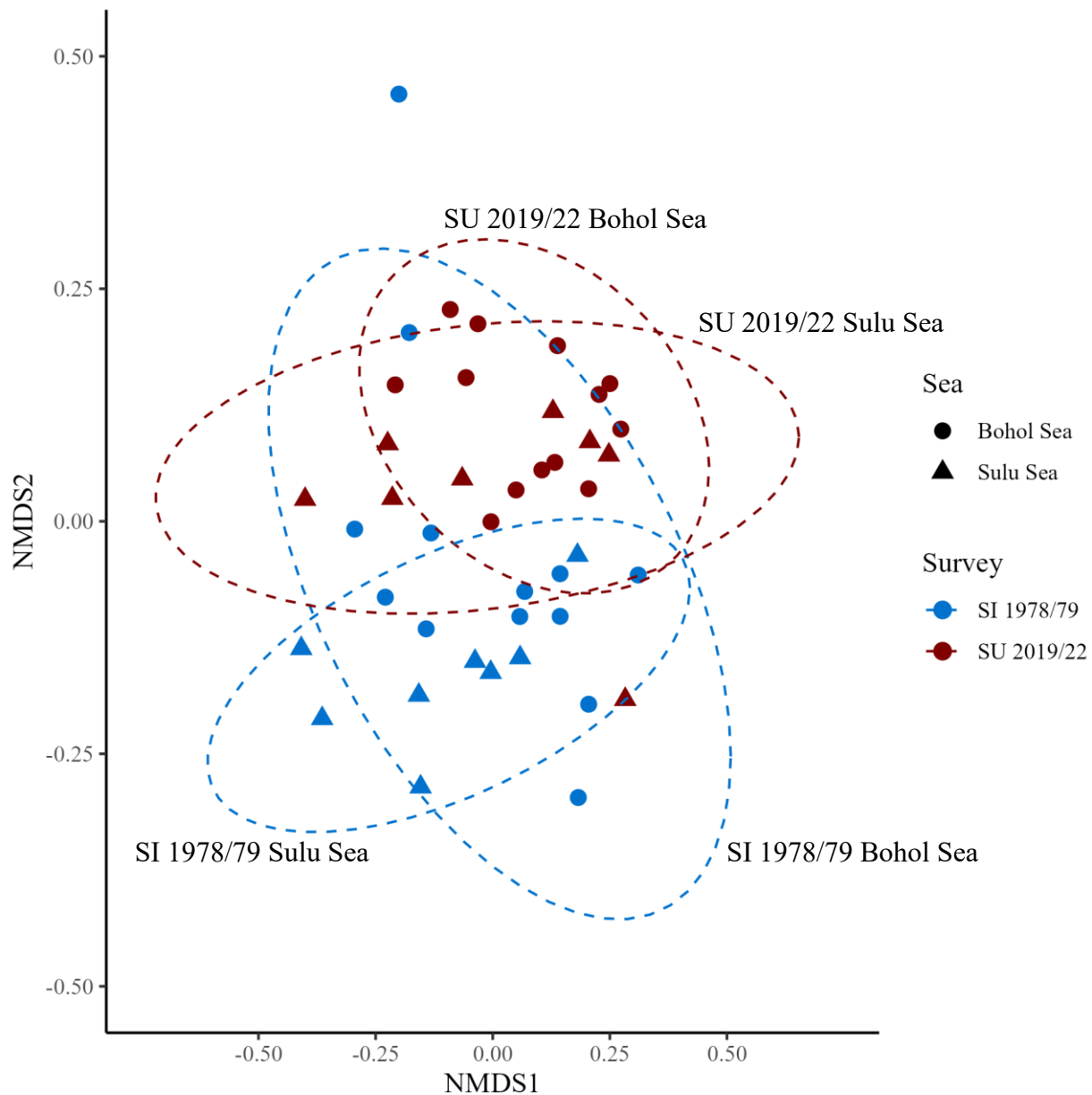


Figure 6. Non-metric multidimensional scaling (NMDS) ordination (Bray–Curtis distances) of reef fish communities surveyed at paired stations in the Sulu and Bohol Seas across two survey eras. Points represent individual stations ($n = 42$), colored by survey era (blue = SI 1978/79, red = SU 2019/22) and shaped by sea (circle = Bohol Sea, triangle = Sulu Sea). Dashed ellipses indicate 95% confidence intervals for each Survey X Sea group.

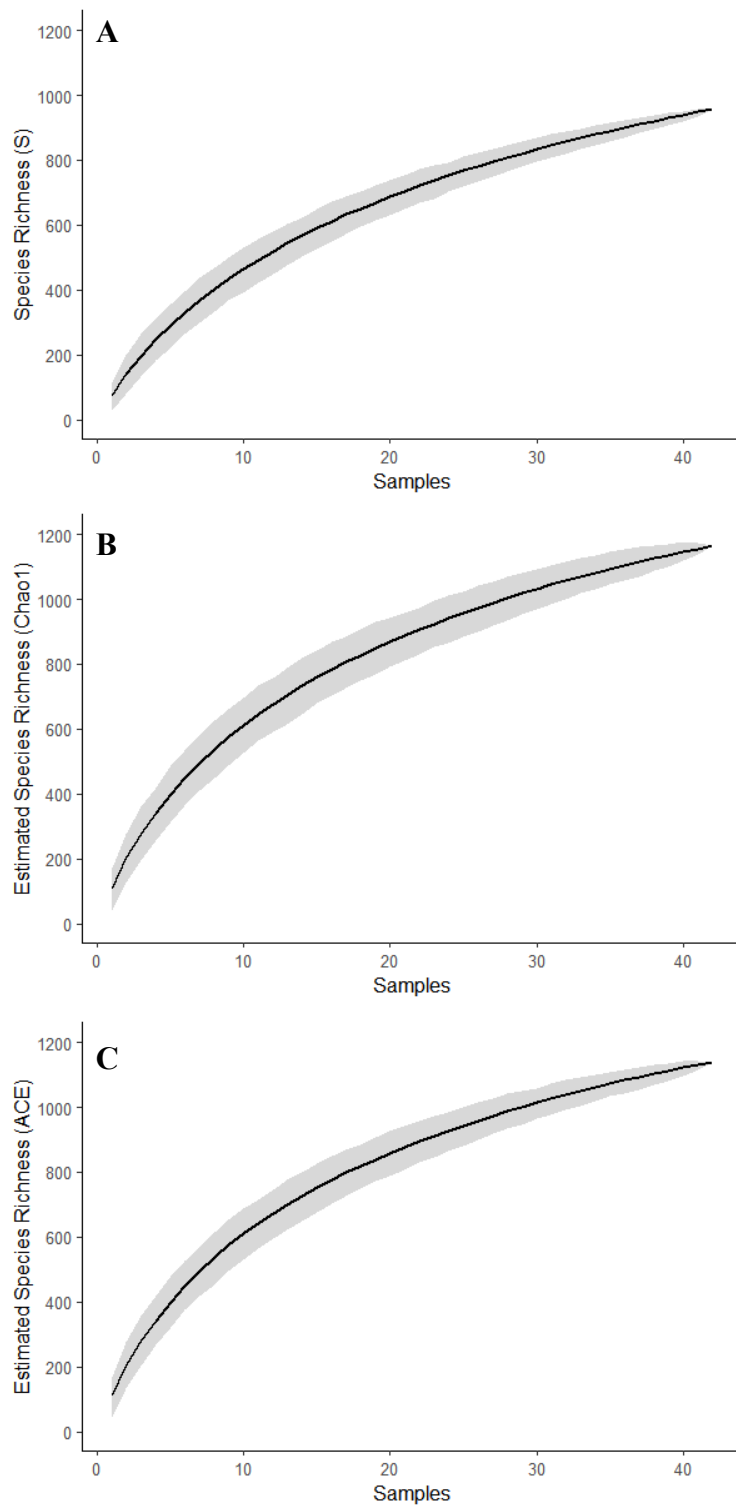


Figure S1. Sample-based species accumulation curves for both survey eras (SI 1978/79, SU 2019/22) and both seas (Bohol & Sulu Sea). Sample sizes for both survey eras was 21 stations for a total of 42. (A) Species richness (S), (B) Chao1 estimator, (C) ACE estimator. Curves show the extrapolated richness indices and number of species for random ordering of sampling units calculated using count data with the *estaccumR* function in the *vegan* package. The ribbons are 95% confidence intervals.

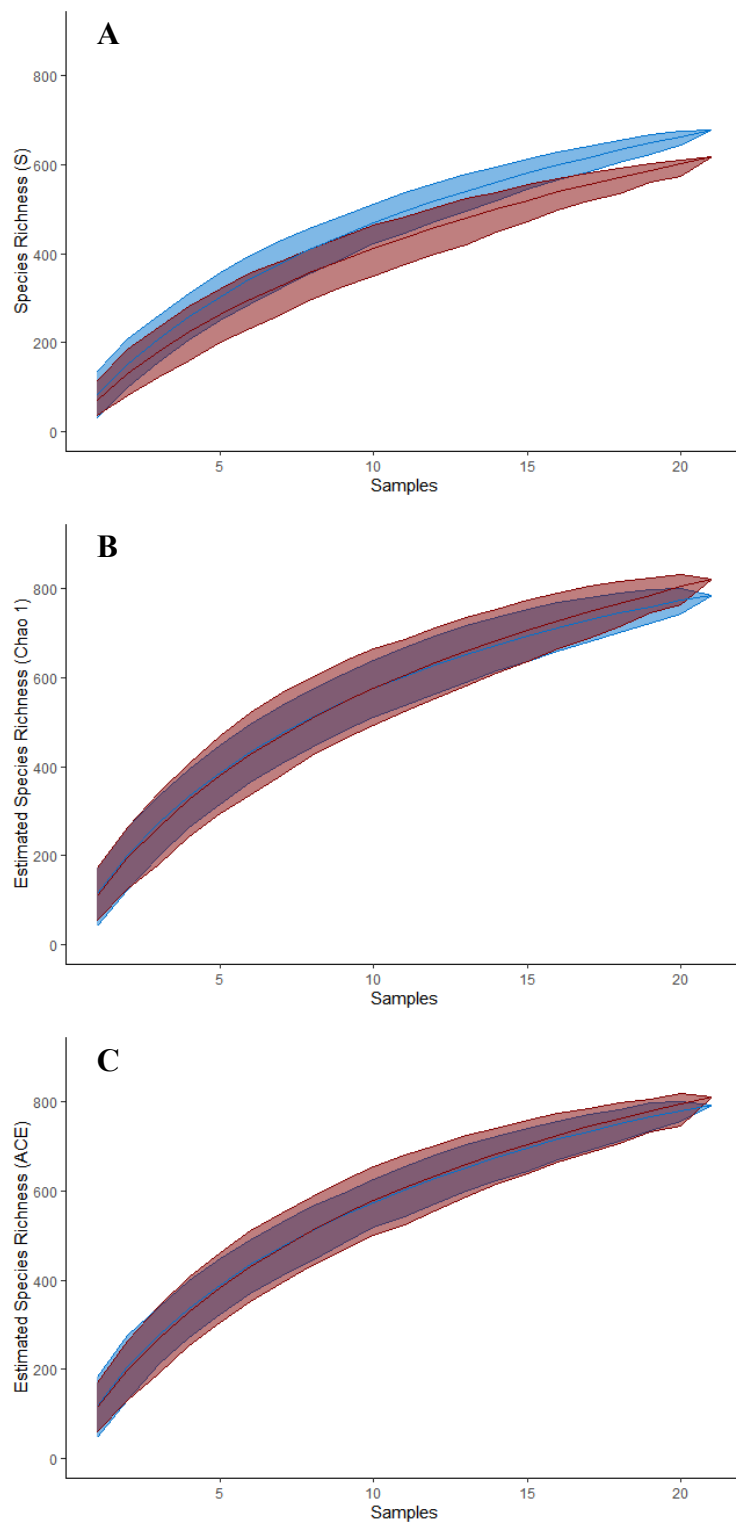


Figure S2. Sample-based species accumulation curves for the Smithsonian Institution survey sampled in 1978 and 1979 (blue) and the Silliman University survey sampled in 2019 and 2022 (red). Sample sizes for both survey eras was 21 stations. (A) Species richness (S), (B) Chao1 estimator, (C) ACE estimator. Curves show the extrapolated richness indices and number of species for random ordering of sampling units calculated using count data with the *estaccumR* function in the *vegan* package. The ribbons are 95% confidence intervals.

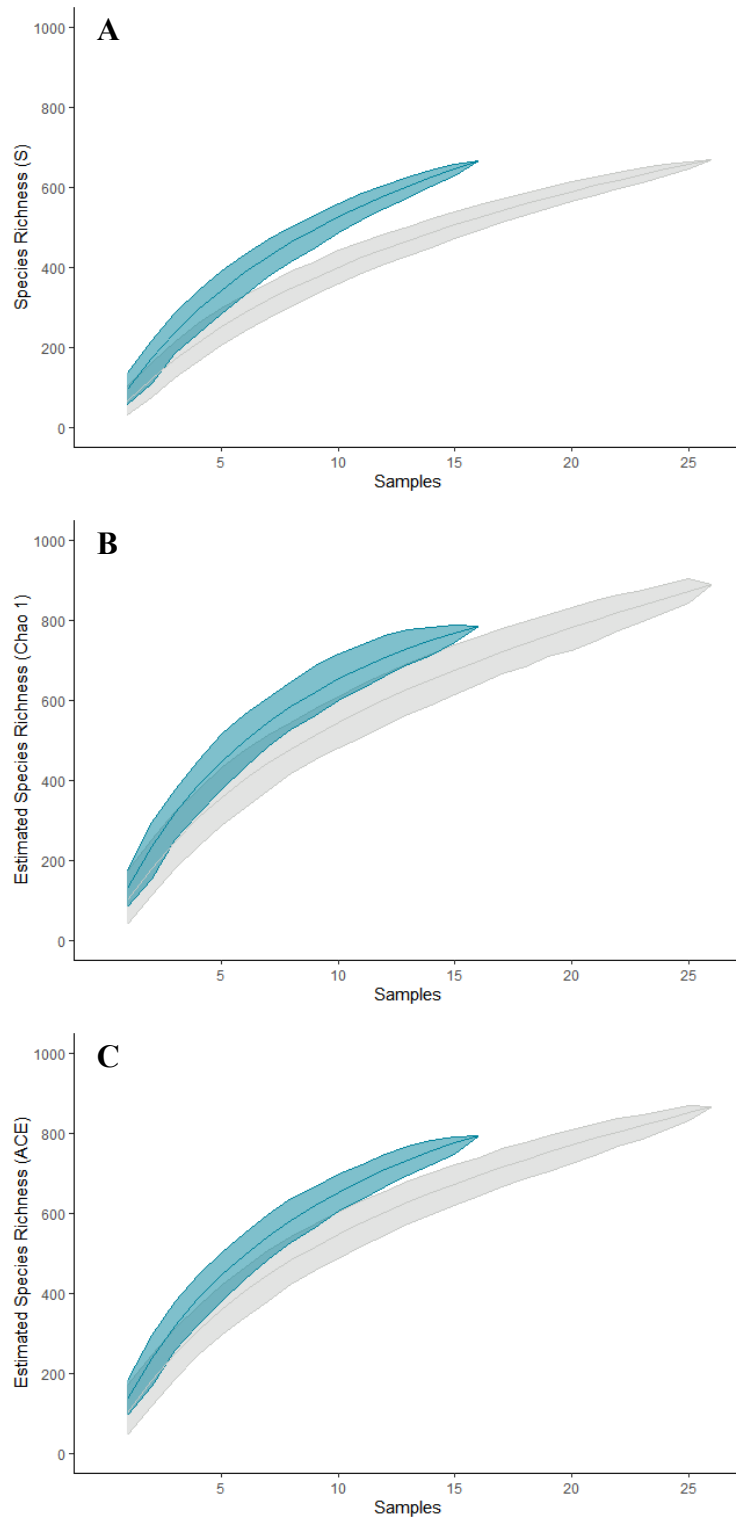


Figure S3. Sample-based species accumulation curves for the Sulu Sea (teal) and the Bohol Sea (grey). There were 16 stations from the Sulu Sea and 26 stations from the Bohol Sea across both survey eras (SI 1978/79, SU 2019/2022). (A) Species richness (S), (B) Chao1 estimator, (C) ACE estimator. Curves show the extrapolated richness indices and number of species for random ordering of sampling units calculated using count data with the *estaccumR* function in the *vegan* package. The ribbons are 95% confidence intervals.

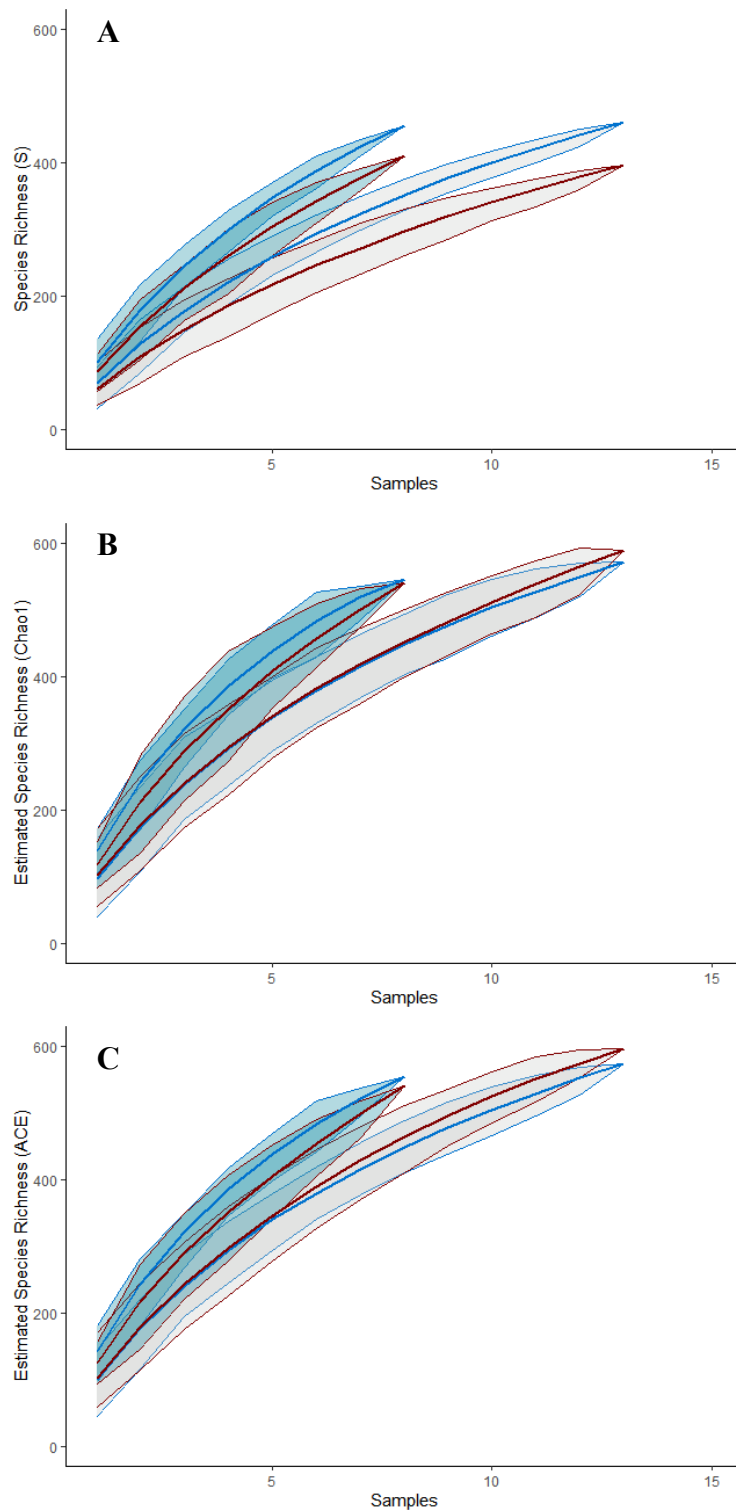


Figure S4. Sample-based species accumulation curves for the Sulu Sea (teal ribbons) and the Bohol Sea (grey ribbons) from the Smithsonian Institution survey sampled in 1978 and 1979 (blue lines) and the Silliman University survey sampled in 2019 and 2022 (red lines). (A) Species richness (S), (B) Chao1 estimator, (C) ACE estimator. Curves show the extrapolated richness indices and number of species for random ordering of sampling units calculated using count data with the *estaccumR* function in the *vegan* package. The ribbons are 95% confidence intervals.

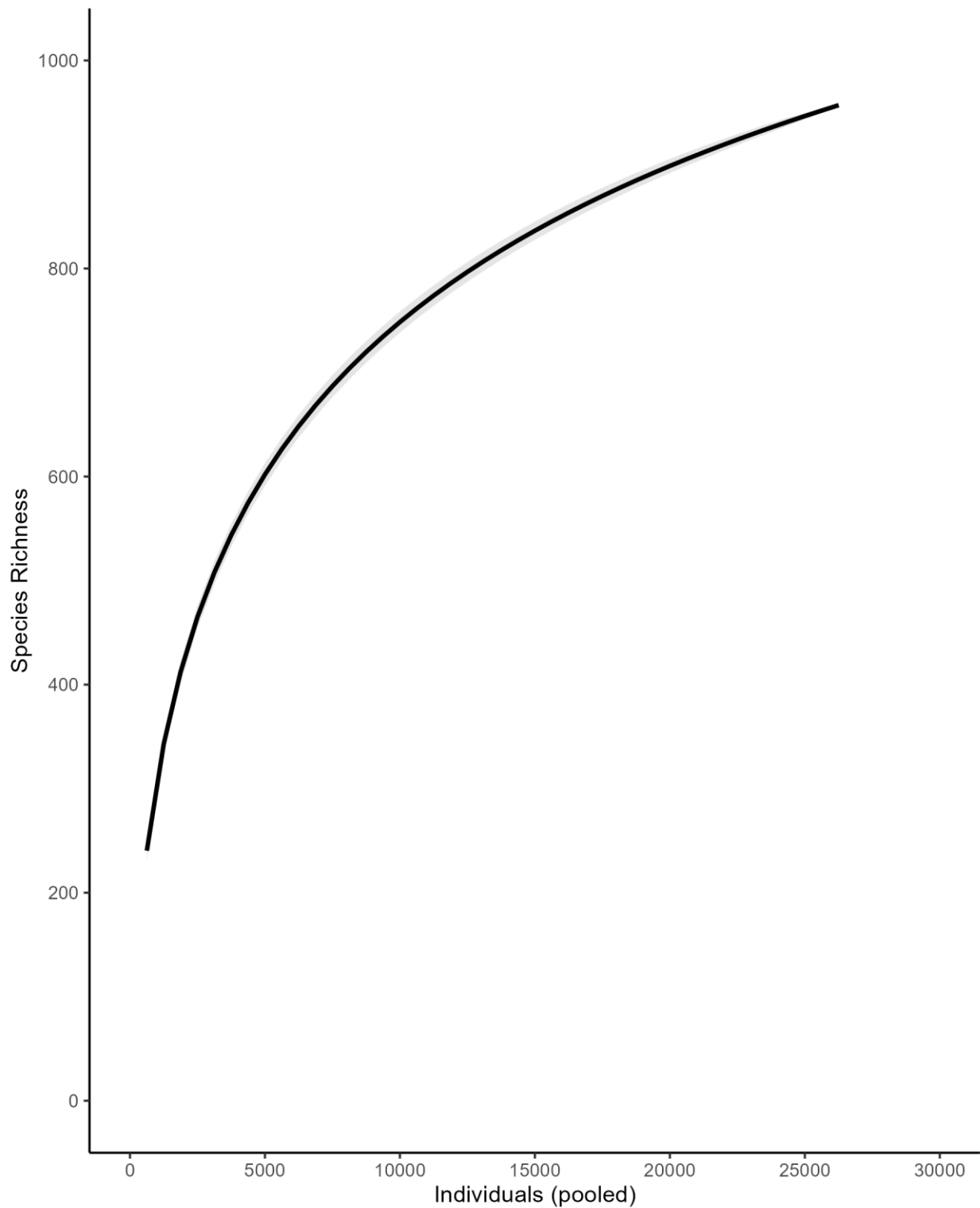


Figure S5. Individual-based species accumulation curves for reef fish communities surveyed in the Sulu and Bohol Seas during both survey eras (SI 1978/79, SU 2019/22). Curves show the expected cumulative number of species as a function of the number of pooled individuals calculated using the rarefaction method in the *specaccum* function of the *vegan* package. Shaded ribbons represent 95% confidence intervals across random permutations.

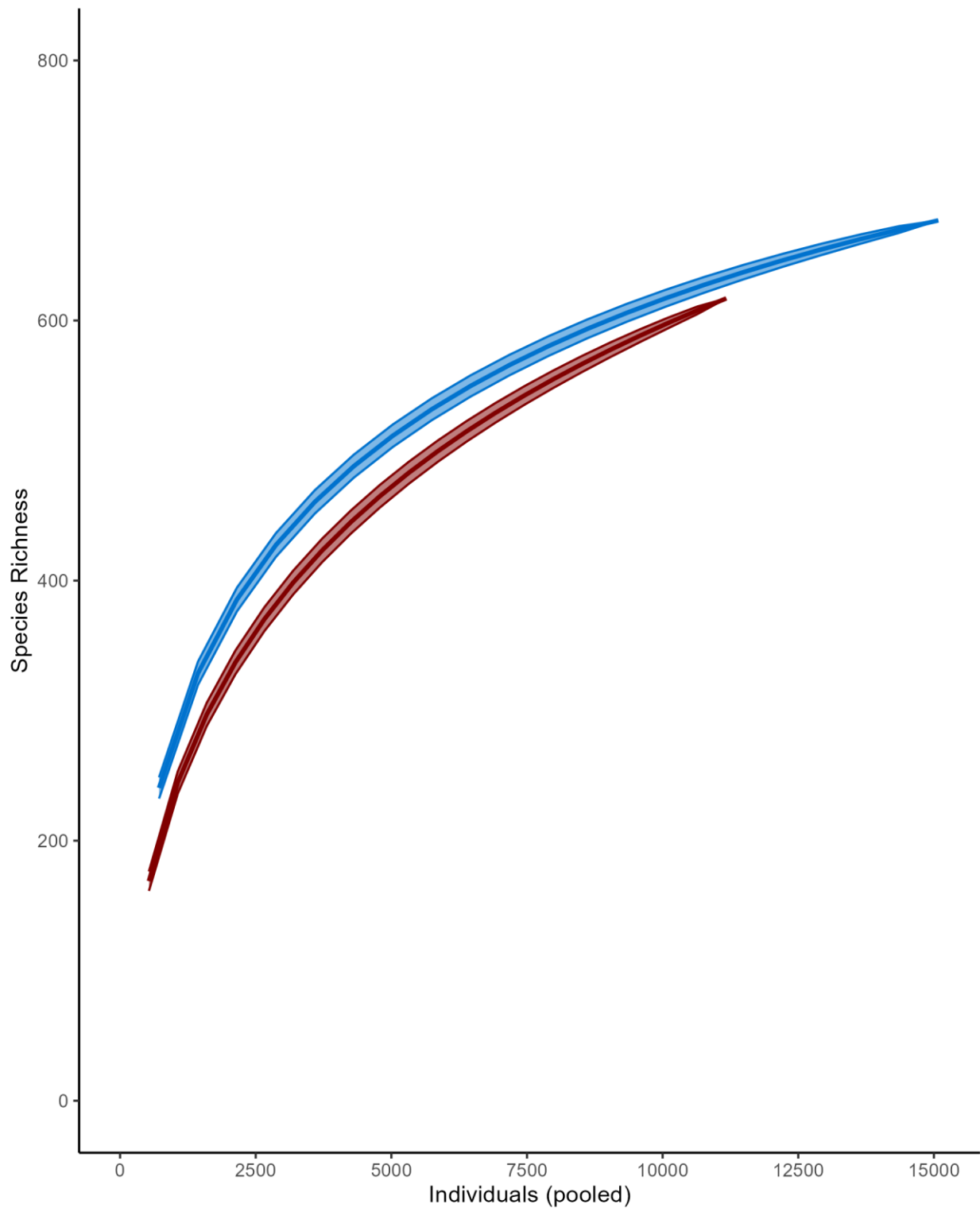


Figure S6. Individual-based species accumulation curves (rarefaction) for reef fish communities sampled during two survey eras (blue: SI 1978/79, red: SU 2019/22) in both the Sulu and Bohol Seas. Curves show the expected cumulative number of species as a function of the number of pooled individuals calculated using the rarefaction method in the *specaccum* function of the *vegan* package. Shaded ribbons represent 95% confidence intervals across random permutations.

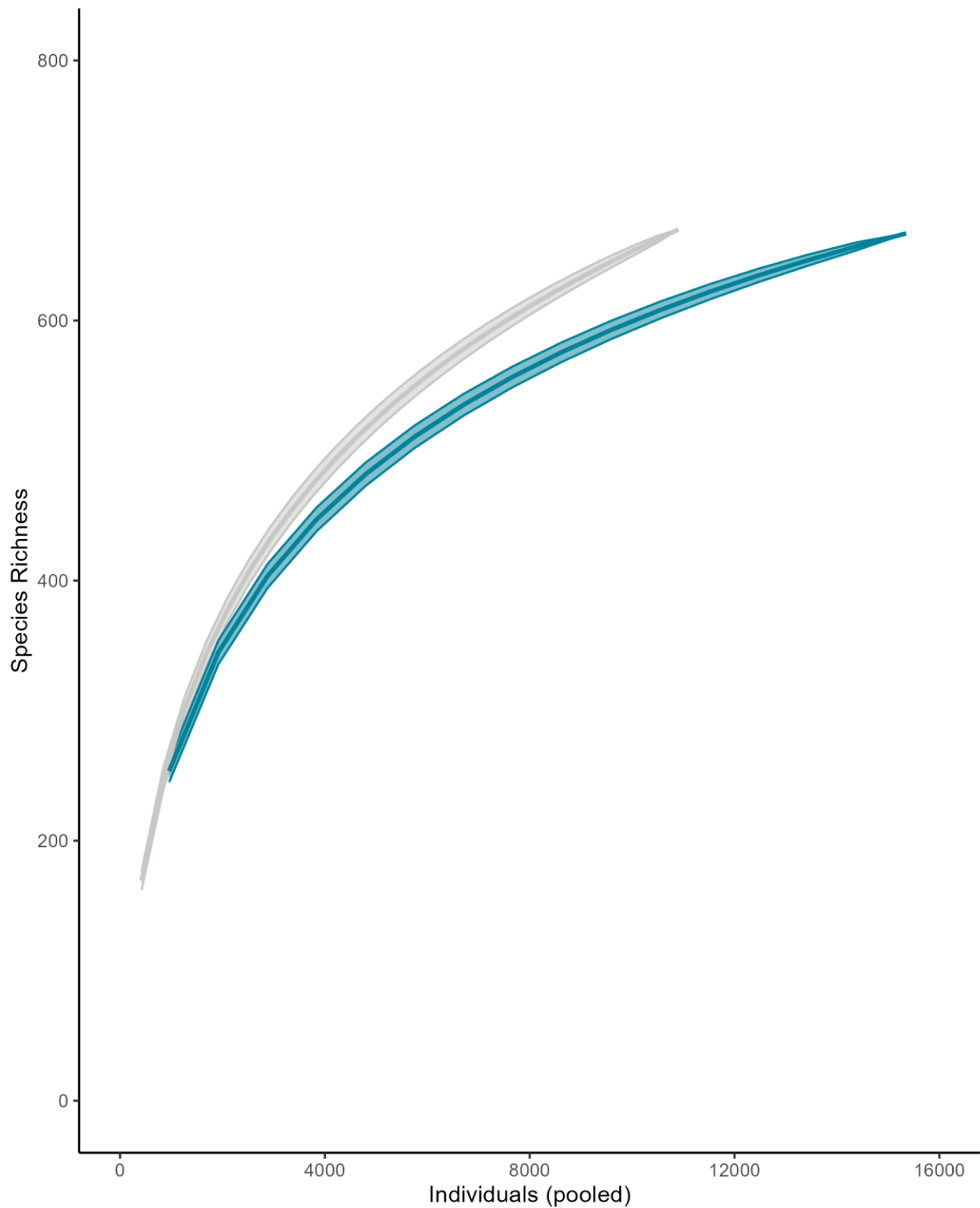


Figure S7. Individual-based species accumulation curves (rarefaction) for reef fish communities surveyed in the Sulu (teal) and Bohol (grey) Seas across two survey eras (SI 1978/79, SU 2019/22). Curves show the expected cumulative number of species as a function of the number of pooled individuals calculated using the rarefaction method in the *specaccum* function of the *vegan* package. Shaded ribbons represent 95% confidence intervals across random permutations.

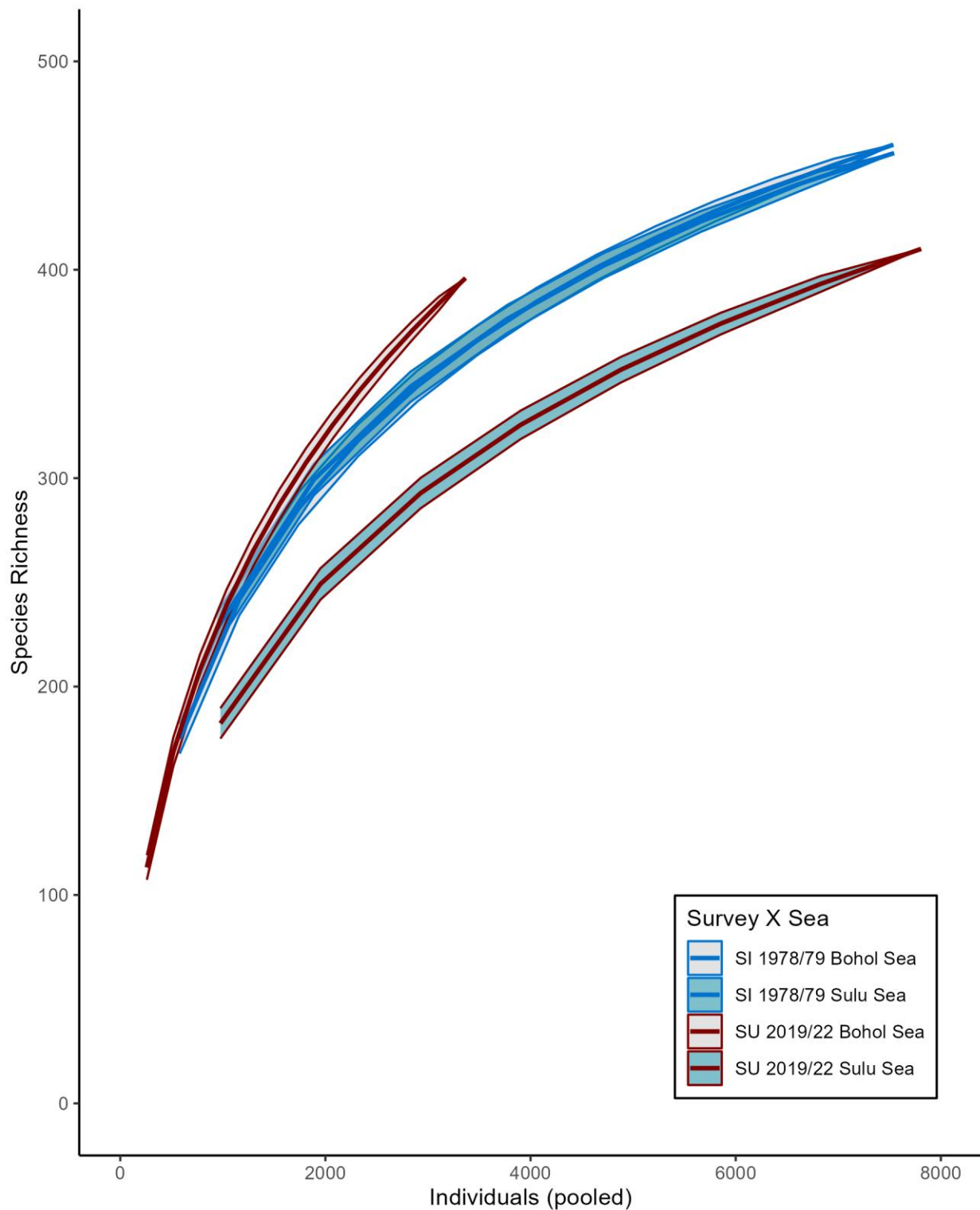


Figure S8. Individual-based species accumulation curves (rarefaction) for reef fish communities surveyed in the Sulu (teal ribbon) and Bohol (grey ribbon) Seas across two survey eras: SI 1978/79 (blue line), SU 2019/22 (red line). Curves show the expected cumulative number of species as a function of the number of pooled individuals calculated using the rarefaction method in the *specaccum* function of the *vegan* package. Shaded ribbons represent 95% confidence intervals across random permutations.